

Product Roadmap (V1 to V4)

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Product: Cubyntra
Organization: Necookie Labs
Current Milestone: V1 (Shipped)

1. Version Overview Matrix

Release	Focus Area	Status	Target Delivery
V1.0	Core Client-Side Vision,	Completed	Q3 2026
V2.0	Validation, Kociemba Solver, 3D	Planned	Q1 2027
V3.0	Web Worker Multithreading, 3D	Planned	Q3 2027
V4.0	Twin Automatic Homography, Bluetooth Smart Cube Support, Warning Voice Cues CFOP Training Suite Physical Robotics Rig, ESP32 Reconstructions Kinematics, Web Serial Controller	Future / Scope Out	2028

2. Milestone Details

2.1 Version 1.0 (Current Release) — The Solid Foundation

- [x] Client-side Computer Vision using WebRTC camera stream.
- [x] Circular kernel color sampling with RGB to HSV and CIELAB transforms.
- [x] ΔE color classification with real-time temporal stability filtering.
- [x] Mathematical invariant validator (cardinality, physical piece integrity, parity).
- [x] Interactive error recovery modal for optical misclassifications.
- [x] Deterministic Herbert Kociemba two-phase solver (< 50ms execution).
- [x] Three.js 3D digital twin with drift-free integer snapping and 3D directional turn arrows.
- [x] Full playback controls (play/pause, step forward/backward, speed multiplier).
- [x] Comprehensive automated test coverage (35/35 passing).
- [x] Complete technical documentation suite and PDF specifications.

2.2 Version 2.0 — Computer Vision & Worker Optimization

- [] Web Workers Solver: Offload Kociemba IDA* search to dedicated background workers, guaranteeing 0ms main thread blocking during complex scrambles.
- [] Automatic Perspective Homography: Automatic four-corner contour detection and perspective rectification, removing the need for a rigid reticle overlay.
- [] Voice-Guided Step Directions: Text-to-speech engine speaking aloud moves (e.g. "Turn the Right face clockwise 90 degrees") for hands-free solving.
- [] Ambient Light Auto-Calibration: Dynamic histogram equalization to handle extreme low-light or yellow tungsten environments.

2.3 Version 3.0 — Smart Cubes & Advanced Speedcubing

- [] Bluetooth Smart Cube Integration: Web Bluetooth API connection to GAN, GoCube, and Moyu smart cubes for real-time hardware telemetry.
- [] CFOP & Roux Method Training: Breakdown of solutions into Cross, F2L, OLL, and PLL stages with algorithmic alternatives.
- [] URL State Serialization: Encode scramble positions and custom solution paths into compact URL hash fragments for zero-backend sharing.
- [] Inspection Timer & Metrics: WCA-compliant 15-second inspection countdown and turns-per-second (TPS) analytics.

2.4 Version 4.0 — Physical Robotics Rig (FUTURE / OUT OF SCOPE FOR V1)

Physical robotics and microcontroller hardware are strictly categorized as FUTURE / OUT OF SCOPE FOR V1.

- [] ESP32 Firmware: Dedicated micro-ROS or lightweight serial firmware for 6-axis NEMA-17 stepper motor coordination.
 - [] Web Serial / Web Bluetooth Execution: Direct hardware bridge from Cubyntra web app to the physical robotic rig.
 - [] Kinematic Acceleration Profiling: S-curve acceleration curves to prevent cube popping during sub-second physical turns.
 - [] Physical Optical Alignment: Dual-camera stereoscopic rig tracking physical cube faces during mechanical rotation.
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